

How WiFi Works

System support

For a system to connect to a WiFi network, it should have a WiFi network card, or an access point connected to it. WiFi network cards are like normal network cards that plug into PCI slots, but have a protruding antenna. It's also possible to plug an external WiFi adapter into a USB port. Once the network cards are installed and configured, the system can connect to any WiFi network. Hardware access points can be connected to network equipment such as a bridge, or it can be connected to a system that has access to a wired network. It's also possible to install software that will convert the system into an access point. Hardware access points are generally used in commercial hot spots, while software access points are used to share an Internet connection in homes and SoHos.

Ad hoc WiFi network

Pros

Given the current situation, WiFi should be adopted when there is a demand for high speeds on the move. WiFi is suited for corporates with several executives roaming around the office with their laptops. Homes looking for a means to share a broadband Internet connection can also consider adopting WiFi. If you're buying a laptop, look for WiFi-enabled devices.

Cons

A WiFi setup is rather expensive when compared to a regular LAN. Also, since WiFi drains batteries much quicker, PDA users should avoid using it—especially if they have a Bluetooth option available. It is difficult to provide security in a WiFi network, as it is difficult to control access to the network.

Devices using WiFi

While WiFi may be widely used for communication between computers, there are PDAs that support it. It's also possible to add WiFi support to a PDA by inserting a WiFi card into the SD (Secure Digital) slot. PDAs such as the Toshiba e750 have inbuilt support for WiFi. In March 2003, Nokia announced plans to build mobile phones that can seamlessly switch between 802.11b and public networks. The major problem with WiFi is the high power consumption, which mobile devices just can't afford.

Access point WiFi network

indicate the task group that proposed the modifications. For example, task group 'b' put forth the 802.11b standard.

A wireless network can work in two modes; namely ad-hoc and base station. In an ad-hoc (also called peer to peer) network, several computers equipped with wireless hardware come together to form a network. All the computers communicate with each other directly. Unless one of the computers participating in the ad-hoc network is connected to a wired network, no system can gain access to a wired network. Three friends meeting in a park and sharing files between their wireless laptops is an example of an ad-hoc network.

The other type of network involves the use of an access point (also called base station). The access point acts as the central point of the network, and is the link between the wired and wireless networks. An access point can be a dedicated hardware system, or just software running on a system. In airports and coffee shops, mainly in the US and Europe, access points are provided so that customers can walk into the shops and start using the Internet from their laptops. It may not be possible to cover an entire airport, or an office, with a single access point. In such cases, more than one

access point is required. Users can take their devices and roam across different access points. All access points have a SSID (Service Set ID) assigned to them, as do all the clients connected to them. When an access point is installed, the default SSID must be changed to increase security, and to prevent users from connecting to some other spot.

The name WiFi was first given to the 802.11b standard. Approved in 1999, 802.11b offered extensions to the original 802.11 that improved the highest data rate from 2 Mbps to 11 Mbps. It operated in the 2.4 GHz frequency range, like as the original and has a range of about 300 feet. Keep in mind that range and speed are interrelated—as the wireless clients move further away from each other, the speed is bound to deteriorate.

802.11a is a standard that operates at a frequency of 5 GHz, and hence it is incompatible with 802.11b. So, a laptop with an 802.11b card can't connect to an access point running on 802.11a. Also, the range of 802.11a is lower than that of 802.11b; hence, more access points are needed to cover a large area. On the brighter side, it offers a much higher throughput of 54 Mbps. Interference is a problem with wireless networks. Bluetooth devices and cord-

less phones are some sources of this interference. 802.11a is less susceptible to interference when compared to other standards such as 802.11b.

To solve the limitation of 'b' and at the same time provide the speed of 'a', the 802.11g was introduced. 802.11g offers speeds of 54 Mbps, while maintaining compatibility with 802.11b networks, so a laptop with a 802.11g card will be able to use a 802.11b access point.

WiFi has a basic level of security provided at the physical level, called WEP (Wired Equivalent Privacy). But vulnerabilities in WEP have already been discovered, and it can't be used as the only security mechanism. However, it is recommended that WEP be turned on, as it provides a basic level of security. Additional levels of security can be provided at the software level in the form of firewalls. The level of security offered by WEP depends on the number of bits used for the encryption key; almost all WiFi products come with a 40-bit encryption key. A 104-bit encryption key is also available, and it is recommended that the largest available key should be used. 

MOULY ARUN PRABHU

mouly_arunprabhu@thinkdigit.com